

B.Tech. Degree III Semester Examination November 2013

ME 1305 METALLURGY AND MATERIAL SCIENCE (2012 Scheme)

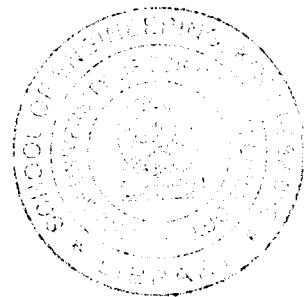
Time : 3 Hours

Maximum Marks : 100

PART A (Answer *ALL* questions)

(8 x 5 = 40)

- I. (a) Calculate the density of BCC iron if its atomic radius is 1.241 Å.
- (b) Metals in general deform by a force which is less than their theoretical strength. Explain.
- (c) Which lead – tin alloy will be ideal for joining electronic assemblies? Give reason.
- (d) Discuss briefly the classification of protective coatings.
- (e) Compare an elastic and visco elastic deformation.
- (f) What is an S-N curve? What useful information is obtained from such a curve?
- (g) What is a stainless steel and why it is stainless?
- (h) Comment on shape memory alloys.



PART B

(4 x 15 = 60)

- II. (a) Obtain the atomic density on (100), (110) and (111) planes of FCC copper. (8)
Lattice constant = 3.61 Å. Identify the slip plane.
- (b) Discuss the Fick's law of diffusion and the metallurgical application of diffusion. (7)
- OR**
- III. (a) Explain the characteristics of dislocations. (5)
- (b) Briefly explain heterogeneous nucleation mechanism. (5)
- (c) Discuss the formation of dendritic pattern during solidification. (5)
- IV. (a) Nickel, Aluminium and Copper have face centered cubic structure yet Ni is soluble in copper whereas Al has only a limited solubility. Explain why it is so. (5)
- (b) Explain eutectic, eutectoid and peritectic reactions. (10)
- OR**
- V. (a) Which microstructure in eutectoid steel has maximum hardness? Give reasons. (5)
- (b) Hardening of steel is always followed by tempering. Is it true or false? Justify your answer. (5)
- (c) What is the need and principle of surface hardening of steels? Explain. (5)
- VI. (a) Discuss any two strengthening mechanisms. (5)
- (b) Explain the difference between recoverable and permanent deformation. (5)
- (c) What is the difference between failure of a material by creep and that by stress rupture? (5)
- OR**
- VII. (a) Discuss the mechanism of fatigue failure. (5)
- (b) What are the different stages in the creep curve? (5)
- (c) What necessitates recovery and recrystallisation processes? How do they influence the properties of materials? (5)
- VIII. (a) Name five alloys each of ferrous and non ferrous metals. Write their composition, properties and applications. (10)
- (b) What is a super alloy? How is it different from stainless steel? (5)
- OR**
- IX. (a) What are properties and applications of various types of cast iron? (10)
- (b) Briefly discuss the applications of titanium and magnesium alloys. (5)